

REPLICABLE AND AUTOMATED MAP AND REPORT PRODUCTION

JONÁŠ GAIGR

Nature Conservation Agency of the Czech Republic
Department of Biodiversity Monitoring



WHAT THIS SESSION COVERS

- **The job** — one protected area, one officer, one question, and the fact that it will be asked again.
- **The two inputs** — occurrence records and thematic layers, and the single step where they meet.
- **The two outputs** — a GeoPackage that opens in QGIS and a PDF that can go in a case file, from one command.
- **What makes it replicable** — configuration in one block, provenance in the file, and the rules that stop it failing quietly.

Everything shown is running in this repository and produced the figures on these slides: github.com/jonasgaigr/kosovo-biodiversity-data-workshop-2026



THE JOB, AND WHY IT REPEATS

Nothing here is a map. It is the machinery that draws the map again next year.



ONE REGISTER, 237 SITES, ONE QUESTION

- Kosovo's nationally designated areas, as the EEA holds them: **237 sites**, 48 with a mapped boundary, 189 recorded as a single point, plus **19 strict-protection zones**.
- Together **1,261 km²**, or **11.6 %** of the country.
- The national report answers national questions. A conservation officer asked about **one site** needs the same evidence for that site alone.

Done by hand, that is: open the register, select the site, clip the records, count the species, draw two maps, look up the Red List and the annexes, write the provenance, export.

Twenty minutes, and it is stale the moment GBIF publishes.

Do it 237 times and the last site is built from a different data snapshot than the first — which is not a reporting method, it is an inconsistency waiting to be found by someone else.



THE TWO INPUTS

	Occurrence data	Thematic data
What it is	Who saw what, where, when	What the ground is, and what the law says about it
Here	46,831 quality-checked GBIF records, 4,095 species	Designated areas, municipalities, national boundary, annex lists, Red List categories
Source	GBIF download service, one DOI	EEA NatDA v24, OpenStreetMap, EUR-Lex, GBIF species API
Changes	Continuously — new records every week	Rarely — a register version, a legal amendment
Geometry	Points, with a stated uncertainty	Polygons, and the occasional point

- They are acquired by different routes, on different clocks, and they are **never edited into each other by hand**.
- Automation is what keeps a fast-moving layer and a slow-moving one in step without anyone remembering to do it.



WHERE THEY MEET: THE STAMP

The pipeline's one join. Every cleaned record is asked which thematic polygons contain it, and the answer is written onto the record as ordinary columns:

Column	From	Rule worth knowing
<code>municipality, district</code>	OpenStreetMap	38 municipalities, 7 districts
<code>protectedArea</code>	EEA register	Where sites overlap, the smallest one is taken — the most specific statement anyone has made about that ground
<code>protectedAreaDesignation</code>	EEA register	In English, from the register's own vocabulary
<code>strictlyProtected</code>	EEA register	Strict zones are excluded from the site test: each lies inside a site already counted
<code>directive, annex</code>	EUR-Lex annex lists	Matched against the GBIF backbone, not by string
<code>iucnRedListCategory</code>	GBIF species API	The Red List — not the IUCN management category of the site

After this step the spatial question is already answered in the attribute table, and every later count is a `filter()`, not a spatial operation.



THE NATIONAL PASS

Rscript pipeline.R

Download → clean → match → stamp → export, in one command. It is **idempotent**: the download key is kept, so re-running does not mint a new DOI and does not re-fetch what is cached.

The first run takes about twenty minutes, most of it resolving common names one taxon at a time. Every later run reuses the download and three caches.

What it leaves on disk, and what everything downstream reads:

- **six thematic subsets** — overall, birds, Birds Annex I, Habitats Directive, Habitats Annex II, IUCN threatened — each as **.gpkg**, **.csv** and, where small enough, **.xlsx**;
- the **protected-area register**, in the same three formats;
- **run_metadata.rds** — the **DOI**, the citation and the cleaning report.



FROM NATIONAL DATA TO ONE SITE

The same evidence, cut to a boundary, in two files a person can actually use.



ONE COMMAND, TWO FILES

```
Rscript site_reports.R --site NP_001
```

```
[1/2] Parku Kombëtar "Sharri" (392171)  
      2,645 records, 1,037 species,  
      parku-kombetar-sharri.gpkg written.  
      parku-kombetar-sharri.pdf rendered.  
      done in 21.7 s
```

- `<slug>.gpkg` — nine layers, the whole evidence base for that site.
- `<slug>.pdf` — a four- to six-page report on **exactly those layers**.
- **No credentials and no network.** It reads what the national pass already published, and never triggers a download.

`--all` for the whole register, `--designation "National Park"` or `--min-area-km2 N` to narrow it, `--gpkg-only` / `--pdf-only` for one of the two, `--force` to rebuild regardless.



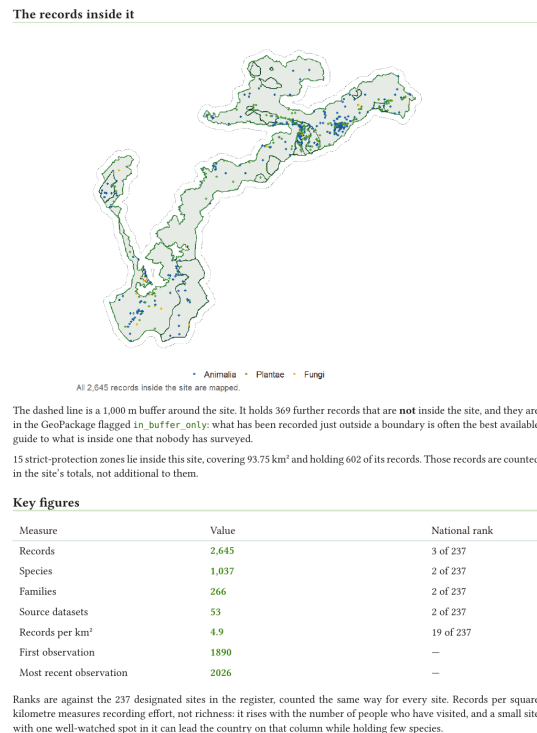
THE GEOPACKAGE: NINE LAYERS

Layer	Geometry	What it holds	Sharri
site_boundary	polygon	the site, and 16 figures computed from it	1
site_buffer	polygon	the 1 km comparison ring	1
strict_protection	polygon	strictly protected cores inside the site	15
occurrences	point	every record in site and buffer, 44 fields	3,014
species_richness_grid	polygon	1 km cells, species per cell	186
record_density_grid	polygon	1 km cells, classed record count	186
municipalities_overlapping	polygon	which units, and what share of each	5
species_summary	—	every species, with Red List and annex	1,037
report_metadata	—	provenance, as name-value pairs	43

Stored in **EPSG:4326**, measured in **EPSG:3035** — both recorded in [report_metadata](#), so neither has to be inferred from the numbers. It opens in QGIS with no conversion and no import step.



THE PDF IS A READER OF THE GEOPACKAGE



- Every figure in the report is **read out of the file beside it**. The report recomputes nothing, so the two cannot state different numbers.
- Ranks are against all 237 sites, counted the same way for every site.
- The dashed ring is the 1 km buffer: 369 further records, flagged `in_buffer_only`. **What is recorded just outside a boundary is the best guide to what is inside one nobody has surveyed.**

Rendered through Quarto's `format: typst`. Typst ships **inside Quarto** — no TeX distribution, no `tinytex`, nothing to install.

IT REPORTS THE GAP, NOT ONLY THE DATA

Species	Common name	Directive	Annex	Records
<i>Picus canus</i>	Grey-Headed Woodpecker	Birds	I	1
<i>Dactylorhiza majalis</i>	Orchide a Fogle Larghe	Habitats	II	6
<i>Euplagia quadripunctaria</i>	Jersey Tiger	Habitats	II	1

25 further listed species are in the *species_summary* layer of the GeoPackage.

Recording effort, and where it is absent

A site with **no records** is **not a site with no species**. It is a site nobody has surveyed, or one whose surveys were never published to GBIF. Everything above describes what has been recorded and published; none of it describes what is there.

The site is covered by 736 grid cells of one square kilometre. 186 of them hold at least one record, and **69 per cent of the site's area falls in cells with no record at all**.

Records run from 1890 to 2026. Of those 137 years, 72 produced no record from this site at all.

Records per cell	Cells	Records
1 – 4	113	210
5 – 19	48	440
20 – 99	18	746
100 – 499	7	1,249

The records come from 53 datasets. The largest contributors are:

Dataset	Publisher	Records	Share
Arthropod observations derived from HabiProt Alciphron database in Serbia	HabiProt	768	29.0%
iNaturalist Research-grade Observations	iNaturalist.org	500	18.9%
EOD – eBird Observation Dataset	Cornell Lab of Ornithology	352	13.3%
Herbarium GZU	University of Graz	263	9.9%
Distributional data on high altitude endemic taxa of beetles (Coleoptera) from neighbouring border areas of western Balkans	Macedonian Ecological Society	209	7.9%

Administrative units

Municipality	District	Area inside the site (km ²)	Share of the site	Share of the unit
Dragash / Dragaš	Prizren	240.07	44.8%	55.3%
Štitërcë / Štrpce	Ferizaj / Uroševac	128.86	24.1%	52.0%
Prizren	Prizren	119.27	22.3%	19.0%
Suharekë / Suva Reka	Prizren	25.17	4.7%	7.0%
Kaçanik / Kačanik	Ferizaj / Uroševac	22.04	4.1%	10.4%

Provenance and citation

Item	Value
Occurrence data	Global Biodiversity Information Facility (GBIF)
GBIF download DOI	10.15468/dl.vzpesq

- **736 cells of one square kilometre; 186 hold a record.** 69 % of the site’s area falls in cells with no record at all.
- Records run 1890–2026; **72 of those 137 years produced nothing** from this site.
- The contributing datasets are named, with their publishers and shares — the report says whose data it is standing on.
- Administrative units are listed with the share of the site *and* the share of the unit, which are different questions.

PROVENANCE TRAVELS WITH THE FILE

Item	Value
Protected-area register	EEA Nationally designated areas (CDDA) v24 (July 2026)
Register licence	CC-BY 4.0
Licences in this extract	CC BY-NC 4.0; CC BY 4.0; CC0 1.0
Generated	2026-09-19T19:14:24Z
Tool	site_reports.R 1.0.0
Source revision	8418bac
Storage CRS	EPSG:4326
Measurement CRS	EPSG:3035

Cite the occurrence data as:

```
GBIF Occurrence Download https://doi.org/10.15468/dl.wzpexq Accessed from R via rgbif (https://github.com/ropensci/rgbif) on 2026-09-05
```

Cite the protected-area boundary as: EEA Nationally designated areas (CDDA), v24 (July 2026), CC-BY 4.0, DOI 10.2909/028003e7-7585-4d69-92fc-7f81e0cc2340.

This document is reproduced by:

```
Rscript site_reports.R --site 392171
```

Every figure above is read from parku-kombetar-sharri.gpkg, which is published alongside this document and holds the records, the species list, the grids and this provenance as GeoPackage layers.

Records are counted by intersection with the site boundary. A record falling inside two overlapping sites is therefore reported in both, which is what a reader of either site's report needs; the national report counts each record once, against the smallest site containing it, so its total for this site can be slightly lower.

report_metadata carries 43 items, and the PDF prints the ones a reader needs:

- the **GBIF DOI** and the citation text;
- the **register version**, its licence and DOI;
- the licences present in this extract;
- the **tool version and git commit** that produced it;
- both CRSs, the buffer and the grid size;
- every input file and its modification time;
- **the command that reproduces the document.**

WHAT MAKES IT REPLICABLE

Not the code. The rules the code keeps.



EVERYTHING COUNTRY-SPECIFIC IN ONE BLOCK

```
config <- list(  
  path_register      = "data/kosovo_protected_areas.gpkg",  
  register_fields    = list(site_name = "site_name", designation = "designation", ...),  
  path_occurrences  = "data_exports/kosovo_overall_biodiversity.gpkg",  
  path_boundary      = "data/kosovo_boundary.gpkg",  
  path_directive_subsets = c("data_exports/kosovo_birds_annex_I.csv", ...),  
  crs_metric         = 3035,  
  buffer_m           = 1000,  
  grid_m             = 1000,  
  map_max_points     = 20000  
)
```

- **register_fields** maps **your** register's column names onto the published schema, so a register that is not a CDDA extract is a list to edit, not code to change.
- Set **path_directive_subsets** to nothing where the Nature Directives do not apply: the report then **says so**, instead of showing an empty annex table as though it were a finding.



RE-RUNNING IS CHEAP, AND SAYS SO

An output is rebuilt when any input is newer than it — and **the code counts as an input**. A changed template with unchanged data regenerates, or the tool would quietly republish last month's layout.

Every run appends to `outputs/run_manifest.csv`: one row per site per run, with the counts, the file sizes, the seconds and the status. **The log is a table, not prose** — so a run can be audited with a query rather than by reading it.

Sites	237
First pass	47.5 min
Second pass, nothing changed	3 s
Median per site	11.7 s
GeoPackages · PDFs	92.6 MB · 23.7 MB

On a laptop, with the network unplugged. **116 MB** is the entire site-level evidence base for the network — no server, no licence, no procurement.



MAPS BUILT TO RENDER IN TEN YEARS

- Drawn with `ggplot2`, and with **no tile service of any kind**. A report that fetched a basemap while rendering would look different next year and would not render at all on a machine with no network.
- The map colours are the **same palette as the website's interactive maps**, so a site drawn here and the same site drawn there are the same green.
- The printed report's colours live in one small Typst file, and the test harness fails if they drift out of step with the website's stylesheet.

Nothing in the figures is fetched, sampled or randomised. The step that builds a site's layers is **pure** — same inputs, same tables — which is what lets a reviewer check a figure by re-running rather than by trusting it.



RULES AGAINST THE QUIET FAILURE

- **Above 20,000 records the point layer is withheld** and the density grid is drawn instead, with the reason printed on the map. Records are never silently thinned.
- **Empty layers are written, not omitted.** A missing layer and an empty layer look the same to a reader in a hurry, and only one of them is honest.
- **An unknown argument stops the run.** A typo that is merely warned about produces a run that looks successful and reports on the wrong sites.
- **One site failing is not the run failing.** The error is logged, the loop continues, the summary lists every failure and the exit status is non-zero.
- **One area column, not two.** The register's own figure is dropped in favour of one computed in EPSG:3035; two areas differing in the third decimal place are worse than one.



THE FINDING THE MACHINE KEEPS PRODUCING

- Of 237 designated sites, **22 hold any record at all**. 215 hold none.
- The 22 hold **21,527 records** between them.
- One wetland of **1.1 km²** holds **13,346** of those — 62 % of everything recorded inside a protected area in Kosovo.

A site with no records is not a site with no species. It is a site nobody has surveyed, or one whose surveys were never published.

Every report says that in as many words, because a reader who takes an empty table for an assessment will reach exactly the wrong conclusion.

Read the other way, those 215 sites are a **survey priority list**, produced at no cost as a by-product of the reporting.



A PILOT, END TO END

1. **QA** — every record placed in its evidence tier.
2. **Gap report** — this tool, on the site and a ring around it.
3. **Field verification** — habitat mapping; standardised bird counts.
4. **Delineation** — along habitats, catchments and visible features, never along grid lines.
5. **Assessment** — Annex III, or the SPA criteria species by species.
6. **Network review** — the site's share of each feature, as a range.
7. **Draft SDF** — and a lessons log, which is the real product.

	Sharri NP	Ligatina e Hencit
Route	habitats, SCI/SAC	birds, SPA
Area	535.5 km ²	1.1 km ²
Records	2,645	13,346
Annex I bird species	18	43
1 km cells with no record	69 %	0 %
IBA	Sara mountain	none

```
Rscript site_reports.R --site NP_001 --site SAC_001
```

Point [path_register](#) at a file of candidate polygons, and the same report shows what a draft boundary leaves out.

THE SAME MACHINERY, POINTED AT THE SDF

A draft Standard Data Form can be a third output, built from the same layers.

SDF section	Source in this system	Filled by
1 · Identification	the register, via register_fields	tool, once a site-code rule is agreed
2 · Location: centre, area, regions	site_boundary in EPSG:3035; region overlays	tool
3.1 · Annex I habitats	a habitat map — none exists yet	expert; left blank
3.2 · Directive species	the directive subsets: a candidate list	tool lists; population and A-D assessment: expert
3.3 · Other important species	species_summary : Red List, Annexes IV, V	tool proposes, expert confirms
4, 6 · Quality, threats, management	field survey and the site's managers	expert; left blank
4.5, 7 · Documentation, map	report_metadata ; site_boundary	tool
5 · Protection status, other sites	overlays with the register and the IBAs	tool

- **A field it cannot compute is written empty and flagged**, like an empty layer. An SDF value is never guessed.
- **No GBIF record reaches the SDF tier on its own**: it can list a species, but the population stays [DD](#) until a survey fills it.



THE SDF SPEAKS ITS OWN CODES: CROSSWALKS

Between the tool's vocabulary and the SDF's sits a **crosswalk** — a table, not code, and the part that costs expert time.

Crosswalk	From → to	Why it is not a lookup
Taxa	GBIF backbone → Natura 2000 species codes	Splits, lumps, synonyms; endemics folded away
Habitats	national or EUNIS types → Annex I codes	Many-to-many: same, narrower, broader, overlapping
Designations	national categories → CDDA codes; IDs → site codes	One category can match several codes
Data quality	survey method and coverage → G M P VP DD	A proposal for national agreement
Bird status	season, behaviour, breeding → p r c w , unit	A rule over several fields, per species
Pressures	field notes → Article 17 pressures and threats	Observed in the field, not in the records

- **Each crosswalk is a versioned table** — source, author and date on every row; one more path in the config block, reviewed like data.
- **An unmatched value stops the draft.** A species dropped because its name had no code is the quiet failure again.
- Built once, they serve every site, every update, and the Article 12 and 17 reports, which use the same codes.



TAKEAWAYS

- **Separate the fast layer from the slow one**, and join them in code. Hand editing is where the inconsistency enters.
- **Publish the data file, not only the document.** The GeoPackage is what someone can work with; the PDF is what they can file.
- **Let the report read the data file**, so the two can never disagree.
- **Put provenance in the output**, not in an email: the DOI, the register version, the commit, and the command that rebuilds it.
- **Configuration in one block** is what makes this a tool for a register rather than a script for one country.
- **Automate the whole path** — the point is not this year's 237 reports, it is that next year's cost one command.



THANK YOU FOR YOUR ATTENTION!

jonasgaigr.github.io/kosovo-biodiversity-data-workshop-2026



NATURE IS OUR HERITAGE
AND THE FUTURE

aopk.gov.cz

